

A Three-Model Ensemble for Antimicrobial Peptide Prediction: Dataset Curation, Model Selection and Web Platform Design

ADAM — Antimicrobial Peptide Database Portal

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Abstract

Antimicrobial peptides (AMPs) are short host-defence molecules of growing interest as candidate therapeutics against drug-resistant bacteria. We describe ADAM, a web platform combining a curated AMP database with an ensemble sequence classifier. Three models are served: a support vector machine over amino-acid composition, a position-specific log-odds profile model, and a fine-tuned protein language model. All three are evaluated under one protocol — stratified five-fold cross-validation on 47,409 unique labelled sequences — and combined by accuracy-weighted voting. We report three methodological findings that materially changed the results. First, label curation mattered more than model choice: resolving 216 contradictory labels and removing duplicate sequences that leaked across cross-validation folds corrected a persistent misclassification that four successive modelling changes had failed to fix. Second, a controlled comparison showed that partially fine-tuning a 333M-parameter language model underperformed the same model used frozen when training data was limited, a capacity effect rather than an implementation fault. Third, two previously published accuracy figures for the platform did not survive re-measurement under a consistent protocol, one of which had never been measured at all.

1. Introduction

Antimicrobial peptides are typically short (5–100 residues), cationic and amphipathic, and act largely by disrupting microbial membranes. Because this mechanism differs from that of conventional small-molecule antibiotics, AMPs are attractive candidates against resistant organisms. Computational screening is valuable because experimental assay of every candidate sequence is impractical.

This work addresses two linked problems. The first is predictive: given a peptide sequence, decide whether it is antimicrobial. The second is practical: deliver that capability as a web platform usable by researchers who are not machine-learning specialists, with results that are interpretable and honestly qualified. We argue that the second problem constrains the first — a model whose reported accuracy cannot be reproduced under a stated protocol should not be presented to users as a number at all.

2. Dataset construction

2.1 Sources

Positive examples are drawn from the ADAM curated peptide database. ADAM annotates antimicrobial activity by sub-type rather than by a single keyword, so selection matches any of antimicrobial, antibacterial, antifungal, antiviral, antiparasitic or anti-Gram activity. Matching only the literal term "antimicrobial" would discard the largest single group, Antibacterial, and mislabel the majority of true positives as negatives.

ADAM curates no negative class, so negatives were retrieved from UniProt Swiss-Prot: reviewed entries of length 5–100 residues, excluding every antimicrobial-family keyword. This is a deliberate asymmetry of evidence. A positive is a demonstrated activity; a negative is only the absence of an annotation.

Table 1. Construction of the labelled dataset from source records to final training set.

Stage	Records	Retained
ADAM database export	57,385	—
Activity matches antimicrobial family	30,467	53.1%
Cleaned, length ≥ 5 residues	29,783	51.9%
UniProt Swiss-Prot negatives	18,451	—
Duplicate sequences removed	−610	—
Conflicting labels resolved to AMP	216	—
Final dataset	47,409	—

Of the 57,385 source records, 21,554 (37.6%) carry no activity annotation and cannot be used as labelled positives. The usable positive set is therefore 29,213 unique sequences, not the raw record count.

2.2 Two defects that were costing accuracy

Label conflicts. 216 sequences appeared in both sets: curated as antimicrobial in ADAM, and also swept into the UniProt negative pool because they lacked the excluded keywords. The classifier was shown the same sequence labelled both ways. We resolve these to AMP on evidential grounds, as stated above. Three resolution policies were measured; resolving to AMP gave the highest accuracy and the lowest variance across folds.

Duplicate leakage. Duplicate sequences survived into the shuffle, so a sequence could appear in both the training and validation side of the same cross-validation fold. The model could then be scored on material it had memorised, inflating the reported accuracy. After de-duplication the final set contains no duplicates and no conflicts.

These curation steps had a larger effect than any modelling change attempted during this work. One benchmark peptide, the insulin B chain, was persistently misclassified as antimicrobial across four successive modelling interventions. Rebuilding the negative set corrected it immediately. The cause was negative-set composition, not classifier capacity.

3. Models

Three classifiers are served. They were chosen to span a deliberate range of complexity, so that the benefit of the most expensive model can be stated against meaningful baselines rather than asserted.

3.1 Support vector machine (SVM)

A radial-basis-function SVM over a 20-dimensional amino-acid composition vector scaled to $[-1, 1]$, with balanced class weighting. Training is capped at a 12,000-sequence stratified subsample because RBF-kernel training cost grows steeply with sample count. This is the cheapest model and returns a decision-function score rather than a probability.

3.2 Position-specific profile model (HMM)

A position-frequency matrix built from AMP sequences over a 30-residue window, converted to log-odds against the background amino-acid distribution, and thresholded at zero. It is a lightweight stand-in for a full profile hidden Markov model and captures positional residue preference that composition features discard.

3.3 Fine-tuned protein language model (ESMC FLM)

FLM denotes Fine-tuned Language Model: the transformer is trained on the task rather than used only as a fixed feature extractor. Sequences are mean-pooled over the attention mask into a fixed-width vector and classified by a two-layer head. The last two transformer blocks are unfrozen together with the head; the remaining blocks stay fixed.

Full fine-tuning of all thirty blocks was measured at 16.2 seconds per sequence on the available CPU-only hardware, which implies approximately 539 hours for the five-fold protocol and is therefore not attainable. Partial fine-tuning of the last two blocks costs about 1.0 second per sequence.

3.4 Ensemble

Each model casts a binary vote weighted by its own cross-validated accuracy, so a stronger model counts for more. The weighted AMP score is the accuracy-weighted share of votes for AMP; a score of at least 0.5 yields an AMP consensus. Scores near 0.5 are reported to the user as genuinely uncertain rather than as confident calls, because that is precisely the region in which the models disagree.

4. Evaluation protocol

All models are evaluated by stratified five-fold cross-validation with a fixed random seed. Stratification preserves the roughly 62/38 class ratio in every fold, which matters for an imbalanced dataset: an unstratified split can produce folds whose class balance differs enough to dominate the measured difference between models.

For the profile model, the profile is rebuilt from each training fold's AMP sequences alone. This is essential and was previously absent. Scoring sequences that contributed to building the profile measures memorisation, not generalisation.

Accuracy is reported as the mean across folds with its standard deviation. We report the spread deliberately: a single train/test split yields a point estimate with no indication of how much of it is split luck, and single-split figures are not comparable with cross-validated ones.

5. Results

Table 2. Cross-validated accuracy of each served model. All figures are stratified five-fold means with standard deviation.

Model	Accuracy	Std. dev.	Evaluated on
SVM	88.37%	$\pm 0.94\%$	12,000
HMM profile	82.69%	$\pm 0.19\%$	47,409
ESMC FLM	in training	—	12,000

5.1 Two figures that did not survive re-measurement

Re-measuring the two classical models under the common protocol changed both published figures.

Table 3. Previously displayed figures against re-measured values.

Model	Displayed	Measured	Basis of the old figure
SVM	90%	88.37%	Single 85/15 split
HMM profile	75%	82.69%	No measurement existed

The SVM discrepancy is split luck. The single reported split corresponded to the most favourable fold: per-fold accuracies were 90.12, 87.33, 88.29, 87.96 and 88.12 percent. Reporting the first of these as the model's accuracy overstates it by 1.75 points, which is nearly twice the standard deviation across folds.

The profile model figure was worse in kind. The training script builds and saves the profile but performs no evaluation whatsoever, so the 75% displayed on the platform and used to weight the model's ensemble vote had no measurement behind it anywhere in the pipeline. Measured properly it scores 82.69%, meaning the model had been under-weighted rather than over-stated. Both the displayed figure and the ensemble weight have been corrected.

We report this because it bears on the paper's own claims. An accuracy figure is only as good as the protocol that produced it, and a figure carried forward without a reproducible protocol should be treated as absent rather than as evidence.

5.2 Does fine-tuning help? A controlled comparison

Comparing the frozen language model against the fine-tuned one as deployed would confound three variables simultaneously: architecture, training-set size, and frozen versus fine-tuned. A controlled pilot was therefore run in which only the last of these varied — identical sequences, identical stratified split, identical evaluation.

Table 4. Controlled frozen-versus-fine-tuned comparison on identical data and splits (4,800 training sequences).

Training strategy	Accuracy	Trainable parameters
Frozen embeddings + SVM head	95.08%	246,530
Last two transformer blocks unfrozen	93.50%	22,376,450

Fine-tuning was 1.58 points worse. The explanation is capacity relative to data, not implementation: 22.1M trainable parameters against 4,800 training examples is roughly 4,600 parameters per example. The signature is diagnostic — training loss fell steadily from 0.4392 to 0.1743 while held-out accuracy declined, and per-class recall diverged to 0.99 on AMP against 0.86 on non-AMP.

This result should not be over-generalised, and we state its limit explicitly. It was measured at 4,800 training examples. At the 9,600 examples per fold used in the current configuration the ratio improves roughly four-fold, to about 2,300 parameters per example, and the overfitting pressure that produced the result is correspondingly weaker. Whether fine-tuning wins at larger sample sizes is an open question that this pilot does not settle; the run now in progress is designed to answer it at matched sample size.

5.3 Benchmark peptides

Four peptides with established status provide a sanity check independent of the cross-validation figures: LL-37 and magainin-2 as positives, and the insulin B chain and glucagon as negatives.

Table 5. Benchmark peptide predictions. The ensemble is correct on all four despite one component model failing.

Peptide	Expected	SVM	HMM	Ensemble
LL-37	AMP	AMP	Non-AMP	AMP
Magainin-2	AMP	AMP	AMP	AMP
Insulin B chain	Non-AMP	Non-AMP	Non-AMP	Non-AMP
Glucagon	Non-AMP	Non-AMP	Non-AMP	Non-AMP

The profile model misclassifies LL-37. This is consistent with its measured per-class behaviour — precision 0.93 but recall 0.78 on the AMP class, meaning it is conservative and misses roughly a fifth of true positives. The ensemble outvotes it and returns the correct call, which is the behaviour that motivates ensembling in the first place.

6. Web platform

The platform is a React single-page application built with Vite, served by a Node/Express backend that delegates inference to a persistent Python service. Keeping the models resident in a long-running process avoids paying model load time on every request, which for the language model is the dominant cost.

6.1 Interface architecture

Only the landing page is bundled eagerly; the remaining routes are split into separately loaded chunks, so a first visit does not download the search, clustering and prediction interfaces before rendering anything. The four model prediction pages were originally near-identical implementations of roughly 308 lines each; they now share one parameterised component and declare only their differences in a configuration object, so a change to prediction behaviour is made once rather than four times.

6.2 Reporting genuine progress

Language-model inference is slow enough that a multi-sequence submission can take minutes. Rather than display an indeterminate spinner, sequences are submitted individually so that each completed response is evidence of real progress. The interface reports completed count against total, extrapolates remaining time from measured elapsed time rather than a fixed estimate, and allows cancellation of a run in flight.

6.3 Communicating uncertainty

The ensemble interface states each model's weight and its basis, and explicitly warns that consensus scores near 50% should be read as uncertain. Presenting a weighted vote without exposing the weights would make a composite number look more authoritative than its components warrant.

6.4 Defects identified during development

Several interface defects were found that were actively withholding information from users rather than merely affecting appearance.

Table 6. Interface defects identified and corrected.

Defect	Consequence for the user
CSS custom property referenced but never defined	Invisible search icon; white text on white background
Database size hard-coded into page text	An incorrect figure published; now read from the server
Base rule declared after its overriding media query	Mobile navigation menu unreachable
Grid track sized 1fr rather than $\min(0, 1fr)$	Result metadata overflowed its container
Unescaped delimiters in exported data	Corrupted CSV downloads
Scroll-triggered reveal with no fallback path	Content permanently invisible if the observer never fired
Export columns hard-coded to model identifiers	Export silently broke when models were renamed

7. Limitations

- Negatives are defined by absence of annotation rather than by demonstrated inactivity. A sequence lacking an antimicrobial annotation in Swiss-Prot may simply never have been assayed, so the negative class is noisier than the positive class.
- The SVM and the language model are trained on a 12,000-sequence subsample for tractability, while the profile model uses all 47,409. Accuracy figures across models are therefore not measured on identical sample sizes.
- All work was performed on CPU-only consumer hardware. This excluded full fine-tuning of the language model and constrained the sample sizes above; the architectural conclusions are conditional on that constraint.
- Sequences longer than 128 residues are truncated. This affects 290 sequences (0.61%), all of them AMPs.
- Accuracy is reported on a roughly 62/38 class distribution that reflects database composition, not the prevalence of antimicrobial activity among peptides in general.

8. Conclusion

We present a three-model ensemble for antimicrobial peptide prediction, delivered as a working web platform and evaluated under a single consistent protocol. The most transferable findings are methodological rather than architectural. Label curation — resolving contradictory annotations and eliminating duplicate sequences that leaked across evaluation folds — produced a larger correction than any change of model. A controlled experiment showed that partially fine-tuning a large protein language model can underperform the same model used frozen when trainable capacity is large relative to the training set, and identified the parameters-per-example ratio as the governing quantity. Finally, re-measuring two previously published accuracy figures under a consistent protocol invalidated both, one of which had never been measured at all. We report these negative and self-correcting results in full, on the view that a prediction platform's credibility rests on the reproducibility of its stated numbers rather than on their magnitude.

Status note: the ESMC FLM cross-validation run was in progress at the time of writing. Its accuracy cell in Table 3 is marked accordingly and must be completed before circulation. All other figures in this document are measured and final.